



Norwegian University of
Science and Technology

FLYT2301: AVIONICS AND FLIGHT SYSTEMS

Lecture 1-1

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17.08.2026

Overview

About the course

Overview

History & Scope of Avionics

The Avionics Subsystems

Aircraft Types Overview

Avionics vs. Aircraft Systems

Architecture & Trends

Regulatory Bodies

Course Roadmap

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About the course

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Centre for Autonomous Marine Operations and Systems

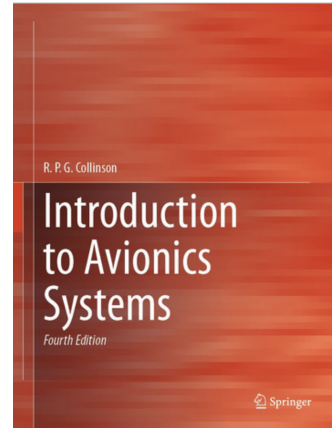
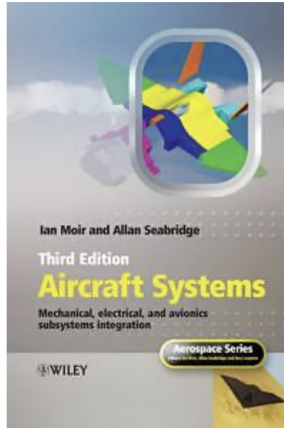
Department of Engineering Cybernetics



Curriculum

Books

- Lecture notes / slides
- Moir, I., & Seabridge, A. (2008). *Aircraft Systems: Mechanical, electrical, and avionics subsystems integration* (3rd). John Wiley & Sons, Ltd. <https://doi.org/10.1002/97804707770931>
- Collinson, R. P. G. (2023). *Introduction to Avionics Systems* (4th). Springer. <https://link.springer.com/book/10.1007/978-3-031-29215-6>
- .. more to come - information will be publish on Canvas



Digital exam

- 4 hours
- More on format later on in the semester
- ... but it will require some detailed knowledge:

Exercises

Exercise Format?

- Format not finalized the
- Some passed will be required to enter the exam
- Some will be voluntary to check learning

Read

- With few exercises use the time well and read the curriculum++
- Great website for additional/complementary info <https://skybrary.aero>
- Learn abbreviations

Excursion: Widerøe Bodø

- 2nd half of the semester
- More info to come
- Primarily for Flyingeniør students
- Additional participants depending on the total number of students registered for the course

Overview



Figure: Aircraft parts: Courtesy: https://www.linkedin.com/posts/shivakant-shukla-a28497270_aircraft-parts-diagram-brief-overview-activity-7319990073079345152-cWLQ

Aircraft Systems: Overview

Topics

- Propulsion/Powerplant
- Flight Controls
- Hydraulic Systems
- Electrical Systems
- Avionics
- Environmental Control
- Pneumatic Systems
- Landing Gear and Brakes
- Fuel Systems
- Ice and Rain Protection

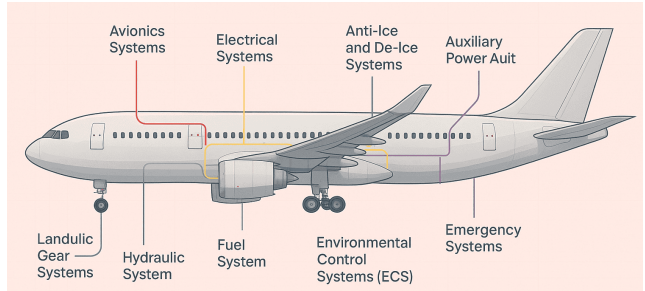


Figure: Aircraft parts: Courtesy:
<https://azazulmahid.com/understanding-aircraft-systems/>

Structure Overview

Config

- 50 % avionics
- 50 % aircraft systems

Platforms cover in the course

Air-v and spacecraft

- Fixed wing aircraft
- Rotary wing aircraft
- Unmanned platforms
- Satellites

Etymology

- “Avionics” is a word derived from the combination of aviation and electronics.
- It was first used in the US in the early 1950s
- Become the industry standard

Avionic system

The term “avionic system” or “avionic sub-system” used to mean any system in an aircraft that is dependent on electronics for its operation (even if the system may contain electro-mechanical elements). Ex:

- Fly-by-wire (FBW) fight control system depends on electronic digital computers for its effective operation, but there are also other equally essential elements in the system.

Propulsion/Powerplant

- engines
- fuel systems, and
- related components for generating thrust.

Flight Controls

- primary (ailerons, elevators, rudder) and
 - secondary (flaps, slats) surfaces, often hydraulically or electrically actuated
- often hydraulically or electrically actuated.

Aircraft Systems (2/5)

Hydraulic Systems

power transmission for

- landing gear,
- brakes
- flight controls, and
- other high-force mechanisms.

Electrical Systems

generation, distribution, and storage of power for

- lights,
- instruments.
- anti-ice/de-ice (for some platforms)
- electronics, and
- other electrical consumers

Aircraft Systems (3/5)

Avionics

Electronic systems for

- communication,
- navigation,
- displays,
- autopilot, and
- monitoring (a subset of overall aircraft systems)

Environmental Control Systems (ECS)

- cabin pressurization,
- air conditioning
- heating, and
- oxygen systems



Aircraft Systems (4/5)

Pneumatic Systems

bleed air from engines for

- anti-icing,
- de-icing, and
- heating,
- other functions

Landing Gear and Brakes

- extension/retraction
- steering, and
- anti-skid braking

Aircraft Systems (5/5)

Fuel Systems

- storage,
- transfer, and
- management of fuel

Ice and Rain Protection

anti-icing and de-icing for

- wings,
- engines, and
- windshield.

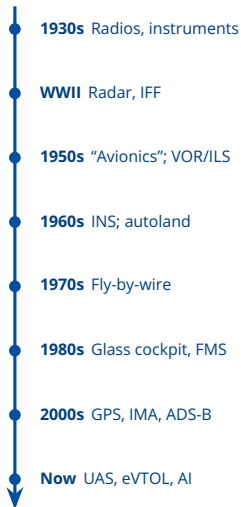


History & Scope of Avionics

A Short History of Avionics

From wires and dials to digital systems

- **1910s–30s:** first radios, instruments, gyros
- **WWII:** radar, radio navigation, IFF
- **1950s:** term “avionics”; analogue autopilots; VOR/ILS
- **1960s:** inertial navigation; Cat III **autoland**
- **1970s:** digital computers; **fly-by-wire**
- **1980s:** **glass cockpit**; FMS; digital FBW (A320)
- **1990s–2000s:** GPS/GNSS; **IMA**; ADS-B; more-electric
- **2010s–now:** UAS, eVTOL, NextGen/SESAR, AI



The Scope of Modern Avionics

What avionics covers

- **Communication** – radios, datalink, SATCOM
- **Navigation** – INS, GNSS, radio aids
- **Surveillance** – transponders, ADS-B, TCAS, weather radar
- **Flight management** – FMS, autopilot, autothrottle
- **Displays & HMI** – glass cockpit, HUD
- **Monitoring** – engine/system health, alerting

Why it matters

- A major and growing share of aircraft **cost** – up to ~30–40% (and higher for advanced military aircraft)
- Central to **safety, efficiency** and **capability**
- Increasingly **software-defined** – functions added by code, not hardware

The CNS + others view

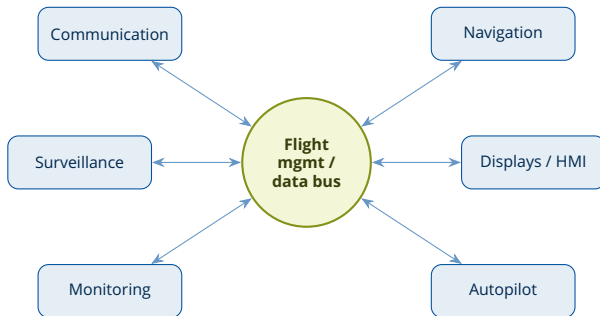
A common grouping: **Communication, Navigation, Surveillance** (CNS), plus flight management, displays and monitoring.

The Avionics Subsystems

An Avionics “Hub”

The integrating idea

- Sensors, radios and computers feed a shared **data network**
- Information is **fused** and presented to the crew (or used by the autopilot)
- Modern avionics is **integrated**, not a box-per-function



Communication, Navigation, Surveillance (CNS)

Communication

Talk & exchange data

- VHF/HF voice
- SATCOM
- Datalink (ACARS, CPDLC)

Navigation

Where am I, where to?

- INS / IRS
- GNSS (GPS, Galileo)
- VOR, DME, ILS

Surveillance

See & be seen

- Transponder (SSR)
- ADS-B
- TCAS, weather radar

The CNS/ATM concept

ICAO's modern air-traffic management is built on **CNS** – precise navigation, reliable datalink communication and cooperative surveillance let more aircraft share the sky safely. (Covered in [Module 3](#).)

Displays, Flight Management & Monitoring

Displays & HMI

- **Glass cockpit**: PFD, ND, multifunction displays
- **Head-up display** (HUD), synthetic/enhanced vision
- Replaces dozens of mechanical instruments

Flight management & guidance

- **FMS** – flight plan, performance, 4D navigation
- **Autopilot / autothrottle** fly the path

Monitoring & alerting

- **Engine/system monitoring** (EICAS / ECAM)
- **Crew alerting** – warnings, cautions, checklists
- **Health monitoring** (trends, maintenance)

The payoff

Together these slashed crew workload – enabling **two-crew** operation of large airliners (down from three or four).

Aircraft Types Overview

Platforms We Will Study

Fixed-wing aircraft

- General aviation, commercial transport, military
- Lift from wings; cruise-dominated; mature, highly regulated avionics
- The **primary focus** of this course

Rotary-wing aircraft

- Helicopters – vertical take-off, hover, low-speed agility
- Extra challenges: rotor control, **vibration**, stability augmentation
- Specialised avionics (e.g. health & usage monitoring, HUMS)

Unmanned aircraft systems (UAS)

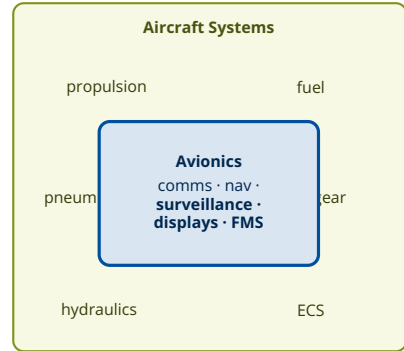
- No on-board pilot – the **autopilot is in full control**
- A **system**: air vehicle + **ground control station** + **C2 datalink**
- Sizes from micro to HALE; rapidly growing civil use

Spacecraft

- Satellites & launch vehicles – no air, no pilot
- **AOCS**, **TT&C**, autonomy, **radiation hardening**
- Avionics principles carry over; environment differs sharply

Avionics vs. Aircraft Systems

Avionics vs. Aircraft Systems



Avionics = the **electronic** subset.

Avionics – the **electronic** systems

- Communication, navigation, surveillance
- Displays, flight management, autopilot
- Monitoring & alerting

Architecture & Trends

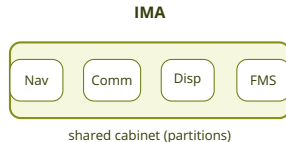
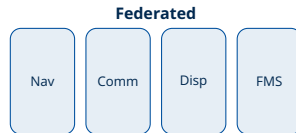
How Avionics Is Organised: Federated → IMA

Federated (traditional)

- One box per function – own processor, power, housing
- Simple to certify in isolation
- But: **heavy**, many spares, poor resource sharing

Integrated Modular Avionics (IMA)

- Shared **computing cabinets** run many functions as **partitioned** software
- Less weight/wiring; easier upgrades
- Needs strong **partitioning** & a robust network (A350, B787)



Returns in depth in [Module 5 \(Digital Avionics & Data Buses\)](#).

A Defining Trend: The More-Electric Aircraft

The shift

Traditionally, four “power” types drove the systems:

- Hydraulic, pneumatic (engine bleed air), mechanical, electrical

The trend is to replace these with electrical power and electric actuation.

Why it matters here

- Pushes more functions into avionics/electronics
- Better efficiency, easier monitoring, fewer fluid systems
- Blurs the avionics / aircraft-systems boundary further

Examples

- B787: no engine bleed air – electric ECS & anti-ice
- Electro-hydrostatic / electro-mechanical actuators for flight controls

Looking ahead

Electric/hybrid propulsion and eVTOL take this even further

Regulatory Bodies

Who Sets the Rules? – ICAO, FAA, EASA

ICAO – global harmonisation

- UN agency (Chicago Convention, 1944)
- Issues **SARPs** (Standards & Recommended Practices) in 19 Annexes
- Sets the global framework – but does **not** certify aircraft

FAA – United States

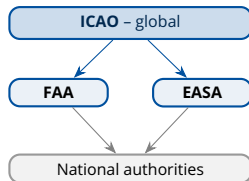
- National regulator; rules in the **FARs** (14 CFR)
- Airworthiness: **Part 23/25** (aeroplanes), **27/29** (rotorcraft); **TSO** for equipment

EASA – European Union

- EU agency; **Certification Specifications CS-23/25/27/29**; Part-21
- National authorities apply the rules (e.g. **Luftfartstilsynet** in Norway)

Why regulation drives avionics

- Defines **safety** and certification requirements
- Mandates equipment (e.g. ADS-B, TCAS)
- Shapes how systems are **designed, built and approved**



Industry Standards Behind the Regulations

Development & safety

- [ARP4754A](#) – system development
- [ARP4761](#) – safety assessment (FHA, FMEA, fault trees)

Software & hardware

- [DO-178C](#) – airborne software
- [DO-254](#) – airborne electronic hardware
- [DO-160](#) – environmental test conditions

Interfaces & data

- [ARINC](#) – equipment & data-bus standards (429, 629, 664)
- [MIL-STD-1553](#) – military data bus

Who writes them

- [RTCA](#) (US) and [EUROCAE](#) (Europe) – the DO-/ED- documents
- [SAE](#) – the ARP documents
- [ARINC](#) – airline-driven equipment standards

These return in detail in [Module 6 \(Certification & Safety\)](#) and [Module 5 \(Digital Avionics & Data Buses\)](#).

Course Roadmap

Lectures

~50% avionics, ~50% other aircraft systems;

Module	Lectures	Key topics	Hours
1 · Intro to Avionics & Systems	L1-L2	History, scope, electrical / hydraulic / pneumatic foundations	4.5 h
2 · Systems Integration	L3-L4	FADEC, IMA, HMI, cockpit architecture	4.5 h
3 · Nav & Comms	L5-L6	GNSS, INS, ILS, ADS-B, TCAS, datalink	4.5 h
4 · Flight Control	L7-L8	Fly-by-wire, autopilot, rotary-wing, emergencies	4.5 h
5 · Digital Avionics	L9-L10	ARINC buses, RTOS, redundancy, FMEA	4.5 h
6 · Certification and Safety	L11-L13	DO-178C/254, ARP4761/4754A, safety culture	4.5 h
7 · UAS & Future	L14-1	UAS, UTM, DAA, eVTOL, AI/ML, cyber	1.5 h
8 · Satellite Avionics and Systems	L14-2	Bus, communication, navigation, ADCS, tasking, payloads, rad-hardening, launch avionics	0.75 h
Total			31.5 h

Course Roadmap: Module 1

Module 1 Introduction to Avionics and Fundamental Systems

3 lectures · 4.5 h

L1 What is Avionics and Aircraft System?

THB

17.08.2026

1×45 min

- History & scope of avionics · Aircraft types overview (fixed-wing, rotary, UAS, spacecraft)
- Avionics vs. aircraft systems · Regulatory bodies (EASA, FAA, ICAO) · Course roadmap

L1-2 Fundamental Aircraft systems part 1

THB

17.08.2026

2×45 min

- Aircraft electrical systems · Power generation & distribution systems · Hydraulic systems

L2 Fundamental Aircraft systems part 2

THB

24.08.2026

3×45 min

- Pneumatic/bleed-air systems · Environmental control systems (ECS) basics · Anti-icing/de-icing

Course Roadmap: Module 2

Module 2 Aircraft Systems Integration

2 lectures · 4.5 h

L3 Engine Control & Propulsion Systems

THB

31.08.2026

3×45 min

- FADEC · Engine sensors & actuators · Fuel systems and Fuel management
- Engine health monitoring · Helicopter turboshaft specifics
- Integration with aircraft systems

L4 System Integration, HMI, SITAW & Cockpit Arch.

THB

07.09.2026

3×45 min

- Integrated modular avionics (IMA) · Cockpit / EFIS · Head-up displays
- Human-machine interface design · Crew alerting systems
- Situational awareness (GPWS/EGPWS, Weather radar, TAWS)



Course Roadmap: Module 3

Module 3 Navigation & Communications

2 lectures · 4.5 h

L5 Navigation Systems

THB

14.09.2026

3×45 min

- Inertial navigation (INS/IRS) · GNSS (and augmentation: SBAS, GBAS)
- VOR, DME, ILS, NDB · Radio altimetry · Barometer
- Area navigation (RNAV) & RNP · Sensor fusion · FMS
- Wide Area Multilateration (WAM)

L6 Communication & Situation awareness Systems

THB

21.09.2026

3×45 min

- VHF/HF/SATCOM voice & data · ACARS · ATC transponders (SSR, ADS-B)
- TCAS/ACAS · Datalink (VDL, LDACS) · Emergency locator transmitters



Course Roadmap: Module 4

Module 4 Flight Control and Emergency Systems

2 lectures · 4.5 h

L7 Flight control systems Part 1

THB

28.09.2026

3×45 min

- Control surfaces · Flight control systems Conventional vs. fly-by-wire (FBW)
- Control laws & flight envelope protection
- Autopilot & architecture · Autothrottle · Autoflight modes · Case studies

L7-2 Flight control systems Part 2

THB

05.10.2026

1×45 min

- Control surfaces · Flight control systems Conventional vs. fly-by-wire (FBW)
- Control laws & flight envelope protection
- Autopilot & architecture · Autothrottle · Autoflight modes · Case studies

L8 Rotary-Wing & Emergency Systems

THB

05.10.2026

2×45 min

- Helicopter flight control specifics · Rotor control systems · Stability augmentation
- Emergency systems (oxygen, fire, ditching, evacuation) · Stall/spin protection · APU

Course Roadmap: Module 5

Module 5 Digital Avionics & Data Buses

2 lectures · 4.5 h

L9 Data Buses & Avionics Computer Architecture

Norse?

12.10.2026

3×45 min

- ARINC 429, 629, AFDX/ARINC 664 · MIL-STD-1553 · CAN bus in aircraft
- Real-time operating systems · Avionics computer hardware · IMA vs. federated architecture

L10 Redundancy, Fault Tolerance & Failure Analysis

THB

19.10.2026

3×45 min

- Redundancy architectures · Failure modes & effects analysis (FMEA) · Fault trees
- Dispatch deviation guides · Built-in test equipment (BITE) · MTBF & reliability concepts

Course Roadmap: Module 6

Module 6 Certification, Safety & Airworthiness

2 lectures · 4.5 h

L11 CAMO and EASA

Wideroe?

26.10.2026

3×45 min

- CAMO · Part Camo · Part 66 · Part 21
- MEL & CS MMEL · Maintenance Steering Group-3 (MSG-3) analyses

L12 Safety Culture, Ethics & Human Factors

THB

02.11.2026

3×45 min

- Just culture & safety management systems (SMS) · Accident investigation (BEA, AAIB methodology)
- Human factors in avionics design · Ethical responsibilities · Case studies

L13 Software, Hardware & System Certification

THB

9.11.2026

3×45 min

- DO-178C (software) · DO-254 (hardware) · ARP4761 (safety assessment)
- ARP4754A (system development) · DAL/SIL levels
- EASA CS-25 / FAA FAR Part 25 · EASA CS-27/29



Course Roadmap: Module 7-8

Module 7 UAS & Future Avionics

1 lecture · 1.5 h

L14 UAS Systems & Emerging Technologies

THB

16.11.2026

2×45 min

- UAS architecture & autopilot · C2 links & UTM · Detect & avoid (DAA)
- UAM/eVTOL avionics · Cybersecurity in avionics · AI/ML in flight systems
- Future regulatory landscape

Module 8 Satellite Systems

1 lecture · 0.75 h

L14-2 Satellite Systems & Spacecraft Avionics

THB

16.11.2026

1×45 min

- Spacecraft avionics architecture · Satellite bus · Attitude & orbit control systems (AOCS)
- Payload systems (EO, SAR, comms) · Ground segment interfaces · Radiation hardening
- GNSS payload systems · Launch vehicle avionics



Summary

Module 1 – Lecture 1: “What is Avionics and Aircraft System? ++”

Today we have introduced:

- What **avionics** and **aircraft systems** are, and how they relate
- A short **history** and the **scope** of modern avionics
- The **platforms** we will study (fixed-wing, rotary, UAS, spacecraft)
- The **regulatory** landscape (ICAO, FAA, EASA) and key standards
- The **roadmap** for the rest of the course

Next – Lecture 2

Electrical, Hydraulic Systems: power generation & distribution

References

References I



Collinson, R. P. G. (2023). *Introduction to Avionics Systems* (4th). Springer.

<https://link.springer.com/book/10.1007/978-3-031-29215-6>



Moir, I., & Seabridge, A. (2008). *Aircraft Systems: Mechanical, electrical, and avionics subsystems integration* (3rd). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9780470770931>

